

SP7022, Qwen3, NPU/GPU, and Machining AI Discussion Summary

Overview

This document summarizes the main topics discussed: SP7022 hardware, LinuxCNC, EtherCAT, NPUs, GPUs, Qwen3 models, machining assistants, tool wear diagnosis, machine vision, and local AI deployment.

SP7022 Hardware

Configuration discussed: SP7022-2331 with Intel Core i5-12400 and 32 GB RAM. Suitable for local AI inference, LinuxCNC, EtherCAT, HMI, and industrial applications.

Using AI on Machine Tools

Potential uses include alarm diagnosis, EtherCAT troubleshooting, maintenance assistance, G-code explanation, process optimization, machine documentation search, and tool wear assessment.

Tool Wear and Vision Systems

A practical architecture is Camera -> Vision Model (e.g., YOLO) -> LLM -> HMI. Vision detects wear or defects, while the LLM explains results and recommends actions.

Recommended Free LLM

Qwen3 was recommended as a strong free model family. Qwen3-8B is a good starting point; Qwen3-14B offers stronger technical reasoning when sufficient RAM is available.

Qwen3 Model Sizes

Qwen3 family discussed: 0.6B, 1.7B, 4B, 8B, 14B, 32B, 30B-A3B MoE, and 235B-A22B MoE.

Recommended Model for SP7022

With 32 GB RAM, Qwen3-14B was recommended as the best balance of quality and resources. Qwen3-8B remains an excellent choice if faster CPU response is preferred.

Running Qwen3 in English

Qwen3 can be configured to respond only in English using a system prompt. No retraining is required to achieve English-only responses.

RAG and Fine-Tuning

Machine manuals, LinuxCNC documentation, EtherCAT references, alarm lists, and maintenance records can be added through Retrieval-Augmented Generation (RAG). LoRA fine-tuning can be

performed later using machine-specific datasets.

NPU vs GPU

GPUs are generally better for Qwen3 because LLMs require large memory capacity, high memory bandwidth, and mature software support. NPUs are often optimized for smaller AI workloads.

When NPUs Are Useful

NPUs can be attractive for machine vision tasks such as tool wear detection, defect detection, and anomaly detection, especially in low-power industrial systems.

Suggested Architecture

SP7022 -> LinuxCNC -> EtherCAT -> Servos. AI layer: Qwen3 + Open WebUI + manuals (RAG).
Optional additions: GPU for faster LLM inference and camera + vision model for tool inspection.

Software Stack

Debian 13, LinuxCNC, EtherCAT, Ollama, Qwen3, Open WebUI, and optional vision models such as YOLO.